

From Games to Learning Agents - A guided tour through Multi-agent Reinforcement learning

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1 Course description

Non-cooperative game theory is a branch of game theory for the resolution of conflicts among players (agents), each behaving selfishly to optimize one's own well-being subject to resource limitations and potentially other restrictions coming from rival's actions. While being a standard modelling technique in economics, the Nash paradigm recently received further attention in connection with adversarial machine learning and multi-agent reinforcement learning. In this course we will introduce the basic tools of these modern techniques. Applications in operations research, economics and analytics will accompany the course.

2 Course timeline

Dates	Topic	Time
9.12.2026	Introduction	9:00-12:00
10.12.2026	Game forms and Solution Methods	10:00-12:00
16.12.2026	Finite Normal form games	10:00-12:00
17.12.2026	Euclidean games	10:00-12:00
18.12.2026	Learning in continuous games	10:00-12:00
21.12.2027	Online learning and regret minimization in games.	10:00-12:00
11.01.2027	Markov decision processes and RL	10:00-12:00
13.01.2027	Markov decision processes and RL	10:00-12:00
19.01.2027	Multi-agent reinforcement learning	10:00-12:00
20.01.2027	Multi-agent reinforcement learning	10:00-12:00
25.01.2027	Multi-agent reinforcement learning	9:00-12:00

3 Assessment

In order to successfully pass the course, the following assesement criteria need to be satisfied:

1. Handing in take home assignments (20 points)
2. Presentation (20 points)
3. Oral exam (60 points)

In total students can reach 100 points.

The grading of the course follows the scheme below: Genügend ≥ 50 , Befriedigend ≥ 60 , Gut ≥ 70 , Sehr gut ab ≥ 80 .

4 Design of the course

The course is a combination of lectures and tutorial style classes. At the beginning of the course, you can select from a list of recent topics in Multi-agent reinforcement learning (MARL). Towards the end of the semester you should be able to present your topic in front of the audience.

5 Learning outcomes

After completing this course, students will know how to model multi-agent and strategic optimization problems in large-scale settings. They will be aware of computational approaches to Nash equilibrium, as well as learning approaches. Modern applications, like multi-agent reinforcement learning will be discussed as well.

6 Recommended literature

There will be self-contained lecture notes made available at the beginning of the course. Complementary readings are:

1. Maschler, Michael, Shmuel Zamir, and Eilon Solan. Game theory. Cambridge University Press, 2020.
2. A Modern Introduction to Online Learning by Francesco Orabona
3. Cesa-Bianchi, N., & Lugosi, G. (2006). Prediction, learning, and games. Cambridge: Cambridge university press.

4. Albrecht, S. V., Christianos, F., & Schäfer, L. (2024). Multi-agent reinforcement learning: Foundations and modern approaches. MIT Press.